

Predicting Ski Jumps using State-Space Model

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Outline

- Introduction
- Modeling Framework and Dataset
 - State-space models and Least squares method
 - Data processing
- Methodology
 - Different approaches and error function
 - Animation app
- Main Results
- Discussion and Conclusion

Motivation

- ▶ Planning of ski jump enlargement and anticipating risky conditions
- ▶ Goal: Predict ski jumps based on external factors

- ▶ State-space models (SSM)
- ▶ Least squares method (LS)

Data Description

- ▶ Data structure
- ▶ Angles
- ▶ coordinate system on a main ski jumping hill in Planica

Angles

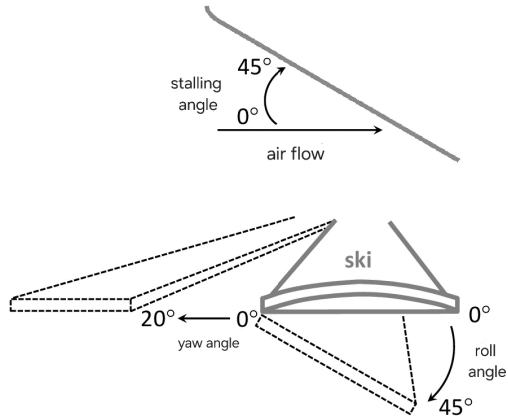


Figure: Representation of different measured angles.

Ski jumping hill

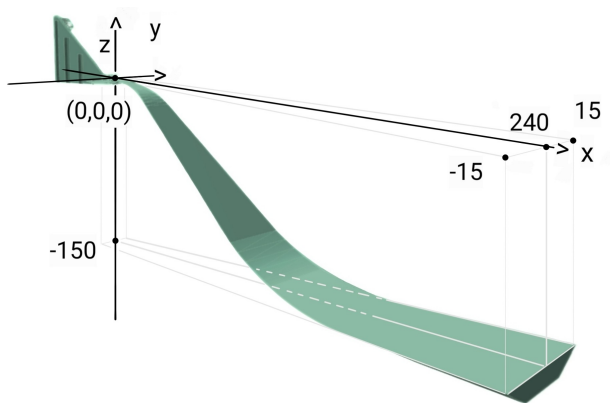


Figure: Ski jumping hill with a coordinate system placed on top.

Approach

- 1 Step one
- 2 Step two
- 3 Step three

Customized caption

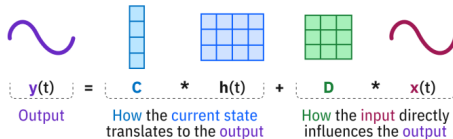
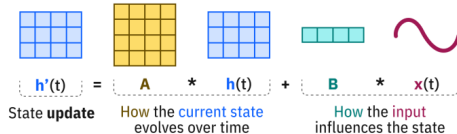


Figure: My first figure in beamer.

Application

- ▶ Description of the application
- ▶ Observing measured and predicted jumps

Main Findings

Key Result

Your important result goes here.

Customized caption

Actual vs Simulated Trajectory
Error: 1.689, Simulated length: 198.7, Actual length: 201.9

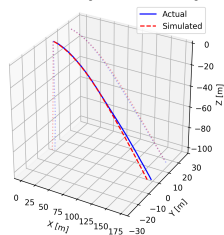


Figure: My first figure in beamer.

Interpretation

- ▶ Limitations
- ▶ Future work and potential improvements
- ▶ Short recap

Thank you!